

# SHARP SELF-CHOLLET INEQUALITIES IN ORDER FIVE: RANK TWO AND STRUCTURED RANK THREE

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ABSTRACT. For a real correlation matrix  $C$ , the self-conjugate specialization of Chollet's conjecture asserts  $\text{per}(C \circ C) \leq \text{per}(C)^2$ . We determine the sharp constant in order five under the rank-two constraint:

$$\text{per}(C \circ C) \leq \frac{137}{1440} \text{per}(C)^2,$$

$$C \in \mathbb{R}^{5 \times 5}, \quad C \succeq 0, \quad \text{diag } C = \mathbf{1}, \quad \text{rank } C \leq 2.$$

Equality holds exactly when the five Gram lines form a regular pentagon in  $\mathbb{RP}^1$ , up to permutation and sign switching. The proof reduces the two permanents to elementary symmetric functions of five unimodular numbers, solves the remaining two-phase minimization, and ends with explicit sum-of-squares and positive-semidefinite Gram certificates. We then study the rank-three stratum. The matrix

$$J_2 \oplus \begin{pmatrix} 1 & -1/2 & -1/2 \\ -1/2 & 1 & -1/2 \\ -1/2 & -1/2 & 1 \end{pmatrix}$$

has self-Chollet ratio  $13/48$ . We prove that this value is sharp in two infinite structured rank-three families and that the displayed configuration is a strict local maximum in the full rank-three configuration space. We also prove Chollet's inequality whenever  $|c_{ij}| \leq 27/50$  for  $i \neq j$ , with the sharper estimate  $\mathcal{R}(C) \leq 621/800$  at coherence  $1/2$ . Finally, without a rank assumption, Chollet's inequality is strict away from the identity whenever  $\lambda_{\min}(C) \geq 3/4$ . The proposed global rank-three value  $13/48$  remains open. All finite algebraic certificates are supplied in exact arithmetic.

## 1. INTRODUCTION

For an  $n \times n$  matrix  $A = (a_{ij})$ , write

$$\text{per } A = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)}.$$

Chollet asked in 1982 whether

$$\text{per}(A \circ B) \leq \text{per}(A) \text{per}(B) \tag{1}$$

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holds for positive semidefinite Hermitian matrices  $A$  and  $B$  [1]. Here  $\circ$  denotes the Hadamard product. In the real symmetric setting it is enough to prove the self-conjugate inequality

$$\text{per}(A \circ A) \leq \text{per}(A)^2. \quad (2)$$

The conjecture was proved in orders at most three by Gregorac and Hentzel [3]. Hutchinson proved the real order-four case [5], and Rodtes subsequently proved the full complex order-four case [8]. Recent work treats parameterized annular classes of complex correlation matrices and, in order five, the ridge class up to permutation similarity [9]; more recent work proves the conjecture for several graph-Laplacian and symmetric  $Z$ -matrix classes [6], but the unrestricted order-five problem remains open. Zhang's analytic study of maximizing correlation matrices provides additional structural background [10].

Rank-two correlation matrices are a natural first boundary stratum. Pate determined the sharp permanent lower bound for real correlation matrices of rank at most two [7]; Drury later recorded the related question whether  $\text{per}(C \circ C) \leq \text{per}(C)$  holds for every real rank-two correlation matrix [2]. The present paper concerns instead the self-Chollet ratio

$$\mathcal{R}(C) = \frac{\text{per}(C \circ C)}{\text{per}(C)^2}$$

and determines its exact maximum in order five.

Our main result is the following.

**Theorem 1.1.** *Let  $C \in \mathbb{R}^{5 \times 5}$  be a correlation matrix with  $\text{rank } C \leq 2$ . Then*

$$\text{per}(C \circ C) \leq \frac{137}{1440} \text{per}(C)^2. \quad (3)$$

*The constant is best possible. Equality holds if and only if, after permuting the indices and changing signs of Gram vectors, the five unoriented Gram lines form a regular pentagon in  $\mathbb{RP}^1$ . Equivalently, an equality representative is*

$$C_{jk} = \cos \frac{(j-k)\pi}{5}, \quad 0 \leq j, k \leq 4. \quad (4)$$

Since  $137/1440 < 1$ , Theorem 1.1 proves a strict strengthening of (2) on this stratum. It also applies, after diagonal normalization, to arbitrary real positive semidefinite  $5 \times 5$  matrices of rank at most two; zero diagonal entries give zero rows and are trivial.

The next natural stratum is rank three. Put

$$C_\star = J_2 \oplus E_3, \quad E_3 = \begin{pmatrix} 1 & -1/2 & -1/2 \\ -1/2 & 1 & -1/2 \\ -1/2 & -1/2 & 1 \end{pmatrix}. \quad (5)$$

Then  $\text{rank } C_\star = 3$  and

$$\text{per } C_\star = 3, \quad \text{per}(C_\star \circ C_\star) = \frac{39}{16}, \quad \mathcal{R}(C_\star) = \frac{13}{48}. \quad (6)$$

Our rank-three results are as follows.

**Theorem 1.2.** *The sharp inequality*

$$\text{per}(C \circ C) \leq \frac{13}{48} \text{per}(C)^2$$

*holds in each of the following rank-three families:*

- (1) *two Gram vectors span a line and the other three lie in its orthogonal plane;*
- (2) *the two-parameter reflection family defined in (35) below.*

*In either family equality is attained precisely, modulo orthogonal transformations, individual sign changes, and relabeling, by  $C_\star$ .*

**Theorem 1.3.** *The configuration  $C_\star$  is a strict local maximizer of  $\mathcal{R}$  in the full space of Gram matrices of five unit vectors in  $\mathbb{R}^3$ , modulo the natural symmetries.*

**Theorem 1.4.** *Let  $C$  be a real  $5 \times 5$  correlation matrix of rank at most three. If  $\max_{i \neq j} |c_{ij}| \leq 27/50$ , then*

$$\mathcal{R}(C) \leq \frac{18981140627902293}{19073486328125000} < 1.$$

*At coherence at most  $1/2$ , one has the sharper bound  $\mathcal{R}(C) \leq 621/800$ .*

These results motivate the sharper global problem

$$\Gamma_{5,3} := \max_{\substack{C \succeq 0, \text{diag } C = \mathbf{1} \\ C \in \mathbb{R}^{5 \times 5}, \text{rank } C \leq 3}} \mathcal{R}(C) \stackrel{?}{=} \frac{13}{48}. \quad (7)$$

The maximum exists by compactness, and  $\text{per } C > 0$  on the feasible set. We do not prove (7); in particular, the structured and local results below must not be read as a global classification.

Our second result removes a full-dimensional neighborhood of the identity from any order-five counterexample search.

**Theorem 1.5.** *Let  $C \in \mathbb{R}^{5 \times 5}$  be a correlation matrix. If  $\lambda_{\min}(C) \geq 3/4$ , then*

$$\text{per}(C \circ C) \leq \text{per}(C)^2.$$

*Equality in this spectral region holds only for  $C = I$ .*

The paper is organized as follows. Section 2 records the normalization, a symmetric-tensor identity, and a two-phase minimization lemma. Sections 3–5 prove Theorem 1.1. Sections 6–8 prove the rank-three results. Section 9 proves Theorem 1.5. The exact finite checks used to audit the displayed expansions are described in Section 10.

## 2. PRELIMINARIES

**2.1. Reduction to correlation matrices.** Suppose first that  $A \succeq 0$  has positive diagonal, and put

$$D = \text{diag}(a_{11}^{-1/2}, \dots, a_{55}^{-1/2}), \quad C = DAD.$$

Then  $C$  is a correlation matrix. For a diagonal matrix  $E$ ,

$$\text{per}(EME) = \left( \prod_i e_i^2 \right) \text{per } M. \quad (8)$$

Applying this once to  $A$  and once to  $A \circ A$  shows that  $\mathcal{R}$  is invariant under positive diagonal congruence. If  $a_{ii} = 0$ , positive semidefiniteness forces the  $i$ th row and column to vanish. This proves the claimed reduction, including all boundary cases.

**2.2. A symmetric-tensor permanent identity.** Let  $v_1, \dots, v_n \in \mathbb{C}^d$ , and write  $\ell_j(z) = \sum_{r=1}^d v_{jr} z_r$ . For a multi-index  $\alpha = (\alpha_1, \dots, \alpha_d)$  with  $|\alpha| = n$ , set  $\binom{n}{\alpha} = n! / (\alpha_1! \cdots \alpha_d!)$ .

**Lemma 2.1.** *If  $G = (\langle v_i, v_j \rangle)_{i,j=1}^n$ , then*

$$\text{per } G = n! \sum_{|\alpha|=n} \frac{|[z^\alpha] \prod_{j=1}^n \ell_j(z)|^2}{\binom{n}{\alpha}}. \quad (9)$$

*Proof.* Let  $P_{\text{sym}}$  be the orthogonal projection from  $(\mathbb{C}^d)^{\otimes n}$  onto its symmetric subspace. Directly from the definition,

$$\text{per } G = n! \|P_{\text{sym}}(v_1 \otimes \cdots \otimes v_n)\|^2.$$

Group the standard tensor basis by multiplicity vector  $\alpha$ . The normalized symmetrization of a word of type  $\alpha$  has norm one, while the coefficient of that vector is the coefficient of  $z^\alpha$  in  $\prod_j \ell_j(z)$  divided by  $\sqrt{\binom{n}{\alpha}}$ . Summing the squared coordinates gives (9).  $\square$

### 2.3. A two-phase minimum.

**Lemma 2.2.** *For  $A, B, C \geq 0$ , the minimum of*

$$A \cos \phi + B \cos \psi + C \cos(\phi - \psi)$$

*over  $(\phi, \psi) \in \mathbb{R}^2$  is one of*

$$B - A - C, \quad -B - |A - C|, \quad (10)$$

*or, when*

$$|A - C| \leq \frac{AC}{B} \leq A + C, \quad (11)$$

*the interior value*

$$-\frac{A^2 B^2 + B^2 C^2 + C^2 A^2}{2ABC}. \quad (12)$$

*The assertion at a zero coefficient is interpreted by continuity.*

*Proof.* Fix  $\psi$ . Minimizing in  $\phi$  gives

$$B \cos \psi - \sqrt{A^2 + C^2 + 2AC \cos \psi}.$$

Thus it remains to minimize the convex function

$$h(z) = Bz - \sqrt{A^2 + C^2 + 2ACz}, \quad -1 \leq z \leq 1.$$

The two endpoint values are (10). An interior critical point satisfies

$$\sqrt{A^2 + C^2 + 2ACz} = \frac{AC}{B}.$$

It exists precisely under (11), and substitution gives (12).  $\square$

### 3. THE RANK-TWO REDUCTION

Choose unit Gram vectors

$$x_j = (\cos \theta_j, \sin \theta_j), \quad 1 \leq j \leq 5,$$

and put

$$\begin{aligned} q_j &= e^{2i\theta_j}, & a &= e_1(q_1, \dots, q_5), \\ b &= e_2(q_1, \dots, q_5), & \varepsilon &= e_5(q_1, \dots, q_5). \end{aligned}$$

Here  $e_k$  denotes the  $k$ th elementary symmetric function. Let

$$u = |a|^2, \quad v = |b|^2, \quad S = 10 + 2u + v.$$

In the complex orthonormal basis of  $\mathbb{C}^2$  adapted to rotations, the coordinate polynomial of  $x_j$  is, up to an irrelevant common convention,

$$2^{-1/2}(q_j^{-1/2}X + q_j^{1/2}Z).$$

Lemma 2.1, together with  $|e_4| = |e_1|$  and  $|e_3| = |e_2|$ , gives

$$P := \text{per } C = \frac{120}{32} \sum_{k=0}^5 \frac{|e_k(q)|^2}{\binom{5}{k}} = \frac{3}{4}(10 + 2u + v) = \frac{3}{4}S. \quad (13)$$

For the Hadamard square, use the Gram vectors  $x_j \otimes x_j$  in  $\text{Sym}^2(\mathbb{C}^2)$ . Their coordinate polynomials in an orthonormal basis are

$$L_j(X, Y, Z) = \frac{q_j^{-1}}{2}X + \frac{1}{\sqrt{2}}Y + \frac{q_j}{2}Z.$$

Consequently Lemma 2.1 gives the directly checkable finite sum

$$Q := \text{per}(C \circ C) = 120 \sum_{r+s+t=5} \frac{|[X^r Y^s Z^t] \prod_{j=1}^5 L_j(X, Y, Z)|^2}{5!/(r!s!t!)}. \quad (14)$$

Using

$$\overline{e_1} = \frac{e_4}{\varepsilon}, \quad \overline{e_2} = \frac{e_3}{\varepsilon},$$

and collecting the twenty-one terms in (14) yields

$$Q = \frac{N}{128}, \quad (15)$$

where

$$N = 685 + 94u + 97v + 21u^2 + 13uv + 4v^2 - 80 \operatorname{Re}(a^2 \bar{b}) - 6 \operatorname{Re}(\bar{\varepsilon} ab^2) - 12 \operatorname{Re}(\varepsilon a \bar{b}^3). \quad (16)$$

Equations (14) and (16) provide two independent descriptions of the same finite expansion.

By (13)–(15), inequality (3) is equivalent to

$$G := 137S^2 - 20N \geq 0. \quad (17)$$

Rotate all  $q_j$  by a common unimodular factor so that  $\varepsilon = 1$ . Write

$$a = xe^{i\alpha}, \quad b = ye^{i\beta}, \quad x, y \geq 0,$$

and set

$$\phi = 2\alpha - \beta, \quad \psi = \alpha + 2\beta.$$

Since  $\phi - \psi = \alpha - 3\beta$ , expansion of (17) gives

$$G = 3600x^2 + 800y^2 + 128x^4 + 288x^2y^2 + 57y^4 + 1600x^2y \cos \phi + 120xy^2 \cos \psi + 240xy^3 \cos(\phi - \psi). \quad (18)$$

#### 4. POSITIVITY OF THE PHASE MINIMUM

Apply Lemma 2.2 to

$$A = 1600x^2y, \quad B = 120xy^2, \quad C = 240xy^3, \quad (19)$$

and introduce the nonnegative variables

$$X = \frac{x}{5}, \quad Y = \frac{y}{10}. \quad (20)$$

We verify nonnegativity on each of the three possible branches.

**4.1. The endpoint  $B - A - C$ .** After division by 10000, expression (18) becomes

$$H_1 = 8X^4 + 72X^2Y^2 - 40X^2Y + 9X^2 - 120XY^3 + 6XY^2 + 57Y^4 + 8Y^2.$$

The identity

$$H_1 = 8(X^2 - Y)^2 + \frac{6}{19}(-2XY - 3Y^2 + 5X)^2 + \frac{21}{19}(-8XY + 7Y^2 + X)^2 \quad (21)$$

proves  $H_1 \geq 0$ .

**4.2. The endpoint  $-B - |A - C|$ .** For  $A \geq C$  and  $C \geq A$ , respectively, the two polynomials obtained after division by 10000 are  $z^\top M_- z$  and  $z^\top M_+ z$ , where

$$z = (X^2, XY, Y^2, X, Y)^\top,$$

$$M_- = \begin{pmatrix} 8 & 0 & -6 & 0 & -2 \\ 0 & 84 & 60 & -18 & 6 \\ -6 & 60 & 57 & -9 & 0 \\ 0 & -18 & -9 & 9 & 0 \\ -2 & 6 & 0 & 0 & 8 \end{pmatrix}, \quad (22)$$

and

$$M_+ = \begin{pmatrix} 8 & 0 & -11 & 0 & -3 \\ 0 & 94 & -60 & 23 & 11 \\ -11 & -60 & 57 & -14 & 0 \\ 0 & 23 & -14 & 9 & 0 \\ -3 & 11 & 0 & 0 & 8 \end{pmatrix}. \quad (23)$$

The successive leading principal minors are

$$(8, 672, 6480, 23328, 0), \quad (8, 752, 2690, 8723, 0). \quad (24)$$

For either matrix the leading  $4 \times 4$  block is therefore positive definite, and the determinant-zero Schur complement is zero. Hence both matrices are positive semidefinite. Their kernels are, respectively,

$$\ker M_- = \text{span}(1, -1, 1, -1, 1)^\top, \quad \ker M_+ = \text{span}(-1, -1, -1, 1, 1)^\top. \quad (25)$$

**4.3. The interior phase minimum.** The phase-dependent contribution in (18) becomes

$$-400x^2 - 1600x^2y^2 - 9y^4.$$

After division by 80000, it remains to prove

$$H_0 = X^4 - 41X^2Y^2 + 6Y^4 + X^2 + Y^2 \geq 0. \quad (26)$$

The upper inequality in (11) is, after (19)–(20), precisely

$$20XY \leq X + 3Y^2. \quad (27)$$

The exact remainder identity

$$H_0 = X^4 + Y^2 + \frac{1}{400} \left( 359X^2 - 246XY^2 + 2031Y^4 \right) - \frac{41}{400} (20XY - X - 3Y^2)(20XY + X + 3Y^2) \quad (28)$$

shows that the last line is nonnegative under (27). The quadratic form in  $(X, Y^2)$  on the first line is positive definite because

$$359 \cdot 2031 - 123^2 = 714000 > 0. \quad (29)$$

Thus  $H_0 \geq 0$ . This completes the proof of (17) and hence the inequality in Theorem 1.1.

## 5. EQUALITY

We now track equality in the preceding certificates. In the interior branch, (28) and (29) force  $X = Y = 0$ . On the two  $-B - |A - C|$  branches, (25) and the nonnegativity of all five entries of  $z$  again force  $X = Y = 0$ .

For the remaining endpoint, the three squares in (21) vanish simultaneously. The first gives  $Y = X^2$ . The second then reduces to

$$X(3X^3 + 2X^2 - 5) = X(X - 1)(3X^2 + 5X + 5) = 0.$$

Thus the only zeros in the nonnegative quadrant are  $(0, 0)$  and  $(1, 1)$ . The latter would require  $|a| = 5$ ,  $|b| = 10$  and  $\arg(a^2\bar{b}) = \pi$ . But  $|a| = 5$  forces equality in the triangle inequality for  $a = \sum_j q_j$ , hence all  $q_j$  coincide. Then  $a = 5q$ ,  $b = 10q^2$ , and  $a^2\bar{b} = 250 > 0$ , a contradiction.

Therefore equality is equivalent to

$$a = b = 0. \quad (30)$$

Since the  $q_j$  are unimodular,  $e_4 = \varepsilon\bar{e}_1$  and  $e_3 = \varepsilon\bar{e}_2$ . Hence (30) gives

$$\prod_{j=1}^5 (z - q_j) = z^5 - \varepsilon.$$

The  $q_j$  are therefore a rotated regular pentagon. Since  $q_j = e^{2i\theta_j}$  records the unoriented Gram line, this is exactly the equality classification in Theorem 1.1. Conversely, for the regular configuration,

$$\text{per } C = \frac{15}{2}, \quad \text{per}(C \circ C) = \frac{685}{128}, \quad \mathcal{R}(C) = \frac{137}{1440}. \quad (31)$$

The proof of Theorem 1.1 is complete.

## 6. STRUCTURED SHARP BOUNDS IN RANK THREE

We begin with a sharp planar lemma. Its elementary form will also identify the equality configuration in (5).

**Lemma 6.1.** *Let  $D$  be a real  $3 \times 3$  correlation matrix with  $\text{rank } D \leq 2$ . Then*

$$\frac{\text{per}(D \circ D)}{\text{per}(D)^2} \leq \frac{13}{24}.$$

*Equality holds precisely when, after sign switching, all three off-diagonal entries of  $D$  equal  $-1/2$ .*

*Proof.* Write the off-diagonal entries as  $x, y, z$ , and put

$$u = x^2, \quad v = y^2, \quad w = z^2, \quad s = u + v + w.$$

The determinant constraint and the frame-potential inequality give

$$1 + 2xyz - s = 0, \quad uvw = \frac{(s-1)^2}{4}, \quad 3 + 2s = \text{tr}(D^2) \geq \frac{9}{2}. \quad (32)$$



Thus  $s \geq 3/4$ . If  $P = \text{per } D$  and  $Q = \text{per}(D \circ D)$ , direct expansion gives

$$P = 2s, \text{quad} Q = 1 + u^2 + v^2 + w^2 + \frac{(s-1)^2}{2}. \quad (33)$$

Writing  $e_2 = uv + uw + vw$ , we obtain

$$13P^2 - 24Q = 4(4s^2 + 12e_2 + 6s - 9). \quad (34)$$

The expression in parentheses is positive when  $s \geq 1$ . When  $3/4 \leq s \leq 1$ , let  $t = ((1-s)/2)^{1/3} \in [0, 1/2]$ . Maclaurin's inequality and (32) yield  $e_2 \geq 3t^4$ , and hence

$$\begin{aligned} 4s^2 + 12e_2 + 6s - 9 &\geq 1 - 28t^3 + 36t^4 + 16t^6 \\ &= (2t-1)^2(4t^4 + 4t^3 + 12t^2 + 4t + 1) \geq 0. \end{aligned}$$

Equality forces  $t = 1/2$  and equality in Maclaurin, so  $u = v = w = 1/4$ . The determinant relation forces  $xyz = -1/8$ , and sign switching gives the asserted representative. The converse is immediate.  $\square$

Suppose now that two Gram vectors span a line and the other three lie in its orthogonal plane. After sign switching and relabeling the Gram matrix is  $J_2 \oplus D$ , with  $D$  as in Lemma 6.1. Permanents factor over direct sums, and

$$\frac{\text{per}(J_2 \circ J_2)}{\text{per}(J_2)^2} = \frac{1}{2}.$$

The first assertion of Theorem 1.2, including sharpness and equality, follows.

We next treat a family that is not block diagonal. Fix an orthonormal basis  $e_0, u, v$  of  $\mathbb{R}^3$ . For  $y, z \in [0, 1]$ , choose  $d \in [-1, 1]$  with  $d^2 = y$  and set

$$\begin{aligned} x_1 &= \sqrt{1-z} e_0 + \sqrt{z} u, & x_2 &= \sqrt{1-z} e_0 - \sqrt{z} u, \\ x_3 &= u, & x_4 &= du + \sqrt{1-d^2} v, \\ & & x_5 &= du - \sqrt{1-d^2} v. \end{aligned} \quad (35)$$

Let  $C(y, z)$  be their Gram matrix.

**Lemma 6.2.** *For the family (35),*

$$48 \text{per}(C \circ C) - 13 \text{per}(C)^2 = -16H(y, z),$$

where

$$\begin{aligned} H &= 15433y^4z^4 - 4336(y^4z^3 + y^3z^4) \\ &\quad + 2424(y^4z^2 + y^2z^4) - 256(y^4z + yz^4) + 64(y^4 + z^4) \\ &\quad + 328y^3z^3 - 36(y^3z^2 + y^2z^3) - 128(y^3z + yz^3) \\ &\quad + 32(y^3 + z^3) - 30y^2z^2 + 48(y^2z + yz^2) \\ &\quad - 12(y^2 + z^2) + 16yz - 4(y + z) + 1. \end{aligned}$$

Moreover,  $H \geq 0$  on  $[0, 1]^2$ , and its only zeros there are  $(1/4, 0)$  and  $(0, 1/4)$ .

*Proof.* The permanent identity is a direct expansion. We give an exact finite positivity certificate for the second assertion. On the boundary,

$$H(y, 0) = (2y + 1)^2(4y - 1)^2, \quad (36)$$

and the analogous identity holds with  $y, z$  interchanged. Near  $(1/4, 0)$  write  $y = 1/4 + \alpha$ ,  $z = \beta$ . Its quadratic part is

$$36\alpha^2 + \frac{225}{32}\beta^2.$$

On  $|\alpha| \leq 1/100$  and  $0 \leq \beta \leq 1/100$ , exact coefficientwise estimation bounds the terms of total degree at least three by

$$\frac{281460096913}{62500000000}(\alpha^2 + \beta^2) < \frac{225}{32}(\alpha^2 + \beta^2).$$

Thus the first rectangle contains no other zero; symmetry treats the second boundary zero.

The complement of these rectangles is partitioned into 23 rational rectangles. Under the affine map of each rectangle to  $[0, 1]^2$ , every coefficient of  $H$  in the bidegree- $(4, 4)$  Bernstein basis is strictly positive. Since Bernstein basis functions are nonnegative and sum to one, this proves positivity on each rectangle. The complete rational endpoints and minimum Bernstein coefficients form part of the exact certificate described in Section 10; an independent verifier reconstructs the polynomial, checks that the rectangles cover the square, and recomputes every coefficient using rational arithmetic. Together with (36), this also proves the zero classification.  $\square$

Lemma 6.2 proves the second assertion of Theorem 1.2. At either zero, the configuration is equivalent to  $C_\star$ ; the equality classification follows.

## 7. LOCAL MAXIMALITY OF THE RANK-THREE CANDIDATE

We prove Theorem 1.3. Work on the configuration space of five labeled unit vectors in  $\mathbb{R}^3$  and put

$$F = 48 \operatorname{per}(C \circ C) - 13 \operatorname{per}(C)^2.$$

At the representative  $C_\star$ , the exact tangent Hessian has characteristic polynomial

$$\lambda^5(\lambda + 414)(\lambda + 432)^2(\lambda + 1638)^2. \quad (37)$$

Three zero directions are generated by the  $SO(3)$  orbit. On an exact seven-dimensional transverse slice, choose coordinates  $(k_0, k_1, w_0, \dots, w_4)$  so that the  $k$ -plane is the remaining Hessian kernel. The five successive leading principal minors of the quadratic form in  $w$  are

$$\begin{aligned} & -2049, \quad \frac{14120379}{4}, \quad -3050001864, \\ & 2939124451536, \quad -1904552644595328. \end{aligned} \quad (38)$$

Sylvester's criterion therefore makes this block negative definite.

Exact Taylor expansion to fourth order gives three further facts: the pure cubic on the  $k$ -plane vanishes; all fifteen coefficients  $w_i k_j k_\ell$  vanish; and the pure kernel quartic is

$$-\frac{225}{2}(k_0^2 + k_1^2)^2. \quad (39)$$

Consequently the Lyapunov–Schmidt effective quartic is unchanged. The analytic implicit-function theorem gives the locally maximizing negative coordinate  $w = w(k)$  and

$$F(k, w(k)) = -\frac{225}{2}|k|^4 + O(|k|^5).$$

Thus  $F < 0$  near, but not on the symmetry orbit of,  $C_\star$ . Since  $\text{per } C > 0$ , this is exactly the claimed strict local maximality of  $\mathcal{R}$ .

## 8. A UNIFORM COHERENCE BOUND

We finish the rank-three analysis by proving Theorem 1.4. Put

$$S = \sum_{1 \leq i < j \leq 5} c_{ij}^2.$$

Since  $\text{rank } C \leq 3$  and  $\text{tr } C = 5$ ,

$$5 + 2S = \text{tr}(C^2) \geq \frac{25}{3}, \quad S \geq \frac{5}{3}.$$

The Grone–Pierce inequality for correlation matrices [4] now gives

$$\text{per } C \geq \frac{1}{5} \|C\|_F^2 = 1 + \frac{2}{5} S \geq \frac{5}{3}. \quad (40)$$

Set  $B = C \circ C$ , and put  $\mu = \max_{i \neq j} |c_{ij}|$ . Every off-diagonal entry of  $B$  belongs to  $[0, \mu^2]$ . The numbers of permutations of five letters with exactly 0, 2, 3, 4, 5 moved points are respectively 1, 10, 20, 45, 44. Therefore

$$\text{per } B \leq 1 + 10\mu^4 + 20\mu^6 + 45\mu^8 + 44\mu^{10}. \quad (41)$$

The right side is increasing in  $\mu$ . Combining (40) and (41) at  $\mu = 27/50$  yields

$$\mathcal{R}(C) \leq \frac{18981140627902293}{19073486328125000} < 1.$$

At  $\mu = 1/2$ , the same calculation gives 621/800.

## 9. A UNIFORM SPECTRAL SHELL

We prove Theorem 1.5. If  $C \neq I$ , write

$$C = (1 - s)I + sG, \quad s = 1 - \lambda_{\min}(C) \in (0, 1/4], \quad (42)$$

where  $G = (g_{ij})$  is a singular correlation matrix. Put  $H = G - I$  and

$$T = \sum_{1 \leq i < j \leq 5} g_{ij}^2. \quad (43)$$

The cycle-cover expansion is

$$\text{per}(I + sH) = 1 + p_2 s^2 + p_3 s^3 + p_4 s^4 + p_5 s^5. \quad (44)$$

There are 10 transpositions, 20 oriented 3-cycles, 15 products of two transpositions, 30 oriented 4-cycles, 20 covers of type  $(2, 3)$ , and 24 oriented 5-cycles. Applying AM–GM to the absolute edge weights, and using  $|g_{ij}| \leq 1$ , gives

$$p_2 = T, \quad |p_3| \leq 2T, \quad |p_4| \leq \frac{9}{2}T, \quad |p_5| \leq \frac{22}{5}T. \quad (45)$$

For completeness, the constants can be read edgewise. Every edge lies in three triangles. Among unoriented 4-cycles it appears six times, and among pairs of disjoint transpositions it appears three times. In the  $(2, 3)$  covers it appears once as the doubled transposition edge and three times as a triangle edge; in the twelve unoriented Hamilton cycles it appears six times. These incidences give exactly (45).

The off-diagonal entries of  $C \circ C$  are  $s^2 g_{ij}^2$ . Thus

$$\text{per}(C \circ C) = 1 + q_2 s^4 + q_3 s^6 + q_4 s^8 + q_5 s^{10}, \quad (46)$$

where all  $q_j \geq 0$ . Applying the same estimates to  $g_{ij}^2$  and using  $\sum_{i < j} g_{ij}^4 \leq T$  gives

$$q_2 \leq T, \quad q_3 \leq 2T, \quad q_4 \leq \frac{9}{2}T, \quad q_5 \leq \frac{22}{5}T. \quad (47)$$

Let  $E = p_2 s^2 + p_3 s^3 + p_4 s^4 + p_5 s^5$ . Since  $E^2 \geq 0$ , equations (45)–(47) yield

$$\begin{aligned} \text{per}(C)^2 - \text{per}(C \circ C) &= 2E + E^2 - (q_2 s^4 + q_3 s^6 + q_4 s^8 + q_5 s^{10}) \\ &\geq T s^2 f(s), \end{aligned} \quad (48)$$

where

$$f(s) = 2 - 4s - 10s^2 - \frac{44}{5}s^3 - 2s^4 - \frac{9}{2}s^6 - \frac{22}{5}s^8. \quad (49)$$

The polynomial  $f$  is strictly decreasing on  $[0, \infty)$ , and

$$f(1/4) = \frac{37441}{163840} > 0. \quad (50)$$

If  $C \neq I$ , then  $T > 0$ ; hence (48) is strict. The identity matrix gives equality, proving Theorem 1.5.

## 10. COMPUTATIONAL REPRODUCIBILITY

The proofs above are analytic; no numerical optimization is used in either theorem. A short exact-arithmetic script performs the following independent finite checks:

- it recomputes (13) directly from the  $5!$  permanent terms and recomputes (16) from (14);
- it derives the three phase-branch polynomials from (18), verifies the sum-of-squares identity (21), and checks the exact  $LDL^\top$  data and kernels for (22)–(23);
- it verifies the remainder identity (28), the cycle counts in Section 9, and the rational value (50).

- for the rank-three results, it recomputes the planar-block identities, the polynomial in Lemma 6.2, the rational Bernstein covering, the Hessian and fourth-order local reduction, and the coherence count in (41); a separate verifier reconstructs the reflection-family certificate without importing its generator.

The scripts use only symbolic integers and rational numbers. Source code, tests, search manifests, and exact certificate data accompany the manuscript. The numerical optimization recorded in the supplement is discovery evidence only and is not used to prove any theorem.

**Data availability.** No external data were used. The complete computational supplement is included with the source distribution.

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